PROSPECTIVE BIKE CUSTOMER ANALYSIS

Identifying Key Customer Segments for Targeted Marketing.

1. INTRODUCTION

Every successful business requires a clear understanding of its market. Entering a market without adequate knowledge of customers and their needs can significantly reduce the chances of business success. While businesses aim to provide valuable products and services to customers, they also seek to generate sustainable returns on investment through profitable operations.

Therefore, effective study and analysis of the market is crucial for business growth. In today's competitive business environment, organizations increasingly rely on customer data to identify key customer segments and make informed strategic decisions. For the bike industry, identifying promising customer segments is essential for developing targeted marketing strategies, allocating resources efficiently, and improving sales performance.

This report presents the analysis of the customer data which helps to answer some business questions and to identify prospective customer segments. The dataset includes customer information across seven key variables: Age Bracket, Region, Occupation, Gender, Marital Status, Education, and Income. These variables are used to segment customers and examine how different characteristics relate to bike purchasing behavior. The segments answer specific questions that contribute to and improve the business decision.

The insights obtained from this analysis are intended to support management in making informed decisions regarding customer targeting, marketing strategies, and future sales initiatives.

2. OBJECTIVES

The primary objective of this report is to identify high-potential customer segments for bike purchases by analyzing customers' demographic, geographic, and socioeconomic characteristics. The analysis seeks to provide data-driven insights that support targeted marketing strategies, improve resource allocation, and enhance overall sales performance.

To achieve this objective, the analysis is guided by three key business questions (the 3Ws):

- What customer characteristics are associated with bike purchasing behaviour?
- Who represents the most promising customer segment for targeted marketing?
- Where are these prospective customers geographically concentrated?

To answer these questions, the analysis aims to:

- Examine the relationship between customer characteristics and bike purchasing behavior.
- Identify the customer segments with the highest likelihood of purchasing a bike.
- Evaluate the influence of age, region, occupation, gender, marital status, education, and income on bike purchasing behavior.
- Generate actionable insights and recommendations to support customer targeting, marketing strategies, and informed business decision-making.

3. OVERVIEW OF DATA

The data set used for this analysis consists of 1,000 customer records and 15 variables (columns). Originally, the dataset contained 13 variables; however, two additional variables were created during the data preparation stage to improve customer segmentation and support a more meaningful analysis. These variables were introduced to better organize the dataset and enhance the identification of prospective customer groups.

Each row in the dataset represents an individual customer, while each column represents a specific variable or feature describing the customer's demographic, geographic, or socioeconomic characteristics. These variables were used to identify purchasing patterns and generate insights into prospective bike customers.

The variables included in the dataset are described below:

- Serial Number: A sequential identifier used to organize the dataset.
- Customer ID: A unique identifier assigned to each customer. This variable distinguishes one customer from another and assists in identifying duplicate records during data preparation.
- Marital Status: Indicates whether a customer is married or single. This variable was analysed to determine its relationship with bike purchasing behaviour.
- Gender: Identifies the customer's gender (Male or Female) and contributes to understanding demographic differences in bike purchasing behaviour.
- Income: Represents the customer's annual income, ranging from 10,000 to 170,000 within the dataset. This variable was analysed to examine the relationship between income level and bike purchases.
- Children: Records the number of children each customer has. Although included in the dataset, this variable was not a primary focus of the current analysis.
- Education: Indicates the highest educational qualification attained by each customer. The dataset classifies education into five categories: Bachelor's Degree, Graduate Degree, High School, Partial College, and Partial High School.

- Occupation: Describes the customer's profession or type of employment. This variable was analysed to understand how occupation influences bike purchasing behaviour.
- Homeowner: Indicates whether a customer owns a home. It provides additional socioeconomic information that may support future analyses.
- Cars: Records the number of cars owned by each customer and serves as an additional socioeconomic indicator.
- Commute Distance: Represents the estimated distance between the customer's residence and workplace. This variable provides insight into customers' commuting patterns and transportation needs.
- Region: Identifies the customer's geographical location. The analysis focuses on three regions: Europe, North America, and Pacific. All regional comparisons and recommendations presented in this report are based on these three regions.
- Age: Records the customer's actual age.
- Age Bracket: Groups customers into three categories; Adolescent, Middle Age, and Old, to simplify the analysis of purchasing behaviour across different age groups.
- Purchased Bike (Target Variable): This is the primary variable in the dataset and forms the basis of the entire analysis. It indicates whether a customer purchased a bike (Yes or No). All other variables were analysed in relation to this target variable to identify customer characteristics associated with bike purchases and to determine prospective customer segments.

4. METHODOLOGY

The methodology adopted for this analysis followed a systematic data analysis process, beginning with data collection and ending with the development of an interactive dashboard for business decision-making. The objective was to transform raw customer data into meaningful insights capable of identifying high-potential customer segments for targeted marketing.

The methodology consisted of the following stages:

4.1 Data Collection

The dataset used for this analysis was obtained from a public GitHub repository and downloaded in Microsoft Excel format. The dataset contains customer records relating to bike purchasing behaviour and includes demographic, geographic, and socioeconomic information. It served as the primary source of data for the analysis and was used strictly for educational and analytical purposes.

4.2 Data Cleaning and Preparation

Before commencing the analysis, the dataset was examined to ensure its accuracy, consistency, and completeness. The data was checked for duplicate records, missing values, and inconsistencies that could affect the reliability of the analysis.

To improve the quality of the analysis, additional variables were created where necessary. Specifically, the Age Bracket variable was generated by grouping customers into three categories (Adolescent, Middle Age, and Old) based on their ages. These transformations simplified customer segmentation and enhanced data interpretation.

4.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted using Microsoft Excel to understand the structure of the dataset and identify trends in customer purchasing behaviour.

Pivot Tables were used to summarize the data and calculate customer counts, purchase rates, and average income across different customer groups. This stage helped identify patterns and relationships that guided the subsequent analysis.

4.4 Data Analysis

The analysis focused on examining the relationship between the target variable (Purchased Bike) and selected customer characteristics, including:

- Age Bracket
- Region
- Occupation
- Gender
- Marital Status
- Education
- Income

Key Performance Indicators (KPIs), including Total Customers, Number of Purchases, Average Income, and Purchase Rate, were calculated to measure business performance. Comparative analyses were then carried out across the different customer groups to identify segments with the highest purchasing potential.

4.5 Data Visualization and Dashboard Development

Interactive charts and dashboards were developed in Microsoft Excel to communicate the analytical findings effectively.

Different chart types were selected based on the nature of the variables being analysed. Interactive slicers for Age Bracket, Region, Occupation, Gender, Marital Status, and Education were incorporated into the dashboard to enable dynamic filtering and facilitate deeper exploration of customer behaviour.

The dashboard was designed to provide management with a concise visual summary of the analysis, enabling quick identification of high-potential customer segments.

4.6 Interpretation and Business Recommendations

The final stage involved interpreting the analytical findings and translating them into practical business recommendations. The insights generated from the analysis were used to identify prospective customer segments, recommend targeted marketing strategies, and support informed business decision-making aimed at improving sales performance.

5. ANALYSIS AND FINDINGS

The analysis was conducted on a dataset containing 1,000 customer records. Of these, 481 customers purchased a bike, representing an overall purchase rate of 48.1%, while 519 customers (51.9%) did not make a purchase. The average customer income was USD 56,360, providing an indication of the general financial profile of the customers. These metrics served as the Key Performance Indicators (KPIs) for evaluating overall customer purchasing behaviour.

5.1 Purchase Behaviour by Age Bracket

Customers were classified into three age groups for the purpose of the analysis, which are; Adolescence, Middle age, and Old.

AGE BRACKET	AGE RANGE
Adolescence	0-30
Middle Age	31-54
Old	54 and above

The analysis revealed that the Middle Age category represented the largest customer segment in the dataset. Out of the 1000 customers analysed, 701 were found to be middle-aged. The old age bracket was observed to be 189 followed by adolescence which recorded a total of 110 sample size.

The Middle Age group recorded the highest number of bike purchases (383) compared with 318 non-purchasers, resulting in a purchase rate of approximately 54.6%. While adolescence has a purchase value of 39 with 71 number of non-buyers indicating a purchase rate of approximately 35.45%.

Old age bracket pulled the highest average income of 59,683 US dollars but has the lowest purchase rate of approximately 31.2% which suggests that income alone cannot determine the purchase behaviour of customers. The Middle Age group recorded a slightly lower average income (USD 59,087) than the old age group (USD 59,683) yet exhibited the highest purchase rate. This finding suggests that while income may influence purchasing behaviour, it is not the sole determinant of bike purchases. Other factors, such as age, occupation, and region, also appear to contribute.

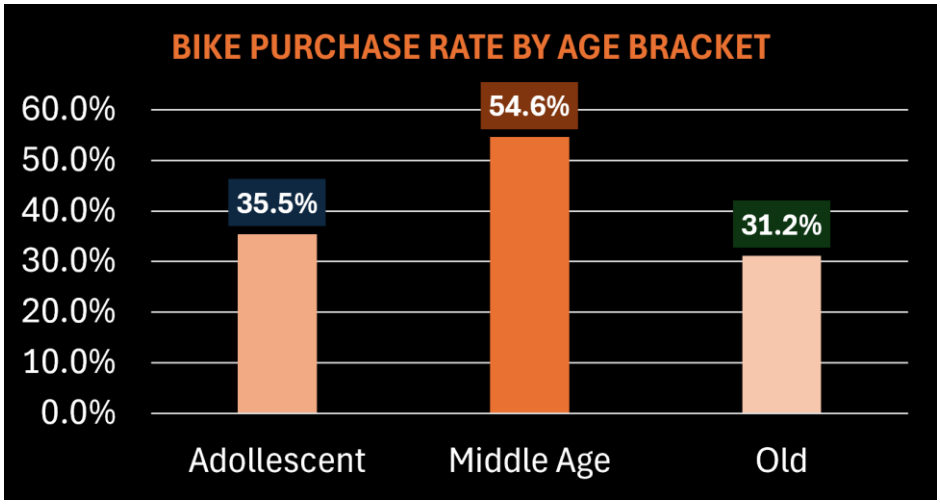


Figure 1: Bike Purchase Rate by Age Bracket

5.2 Purchase Behaviour by Region

North America has 508 recorded customers with 220 and 288 bike buyers and non-buyers respectively. Europe accounts for 30% of the customers in the dataset (300 data points) and has 148 bike purchases and 158 non-purchase bikes. While Pacific has 192 measured data samples with 113 bike purchases and 79 non-purchases.

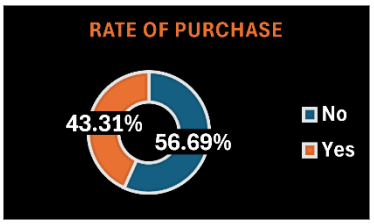


Figure 2: North America Purchase Rate

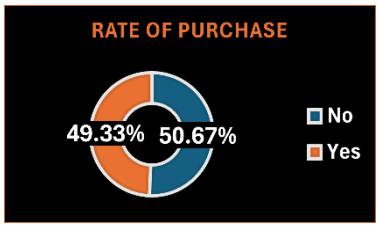


Figure 3: Europe Purchase Rate

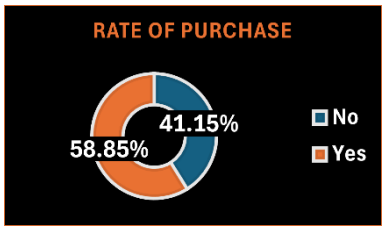


Figure 4: Pacific Purchase Rate

Figure 2, 3, and 4 respectively, shows the purchase rate of the various region segments. The pie chart shows that despite that Pacific has the lowest data sample size, yet it accounted for the highest purchase rate of 58.85% approximately, having Europe and North America be the second and third respectively.

Pacific recorded the highest average income of 63,594 US dollars. This finding indicates that income may contribute to bike purchasing behaviour. Europe has an average income of

40,900 US dollars, recording the lowest mean income level compared to North America with average income of 62,756 US dollars. The result of the average income of customers in Europe shows that income alone does not determine the purchasing power of prospective customers.

5.3 Integrated Customer Analysis

Having established that the Middle Age group played a significant role in determining the characteristics of prospective customers, a deeper analysis was conducted to examine how other customer characteristics, including Region, Occupation, Income, Gender, Marital Status, and Education, interacted with age to influence bike purchasing behaviour.

The analysis revealed that the purchase rates of the Middle Age group in Europe, North America, and Pacific were 56.95%, 51.45%, and 59.09%, respectively. Although the Pacific region recorded the highest purchase rate, Europe and North America also demonstrated considerable market potential among middle-aged customers.

Further examination of the average income across the three regions showed that the Pacific region recorded the highest average income despite having the smallest sample size. In contrast, North America contained the largest proportion of middle-aged customers but recorded the lowest purchase rate among the three regions. Europe recorded an average income of USD 36,816, which was below the overall average income of customers who purchased bikes. Despite this, the region still exhibited a relatively high purchase rate. This finding suggests that income alone does not fully explain customers' purchasing behaviour, and that other customer characteristics may also influence purchasing decisions.

To further refine the analysis, the Middle Age group was examined based on occupation. The results showed that Clerical workers recorded an average income of USD 33,214 and a purchase rate of 57.14%. Although their average income was below the overall average income of bike purchasers, they still demonstrated a relatively high purchase rate. This suggests that occupation may influence purchasing behaviour beyond income level alone.

Similarly, Manual workers recorded the lowest average income (USD 17,000) yet achieved a purchase rate of 46.67%. Although this purchase rate was lower than that of customers who did not purchase bikes within the same occupational group, it indicates that lower income does not necessarily prevent bike purchases. Consequently, occupation should be considered alongside income when identifying prospective customer segments.

Higher-income occupations also demonstrated favorable purchasing behaviour. Management workers recorded an average income of USD 102,073 with a purchase rate of 56.10%, while Professional workers recorded an average income of USD 76,384 and a purchase rate of 58.04%. Skilled Manual workers also exhibited a relatively strong purchase rate of 52.33%, with an average income of USD 55,389. These findings indicate that

occupations associated with moderate to high income generally exhibited stronger purchasing behaviour, although income was not the sole determining factor.

Additional customer characteristics, including Gender, Marital Status, and Education, were also explored during the analysis. However, these variables did not exhibit patterns as pronounced as those observed for Age, Region, Occupation, and Income. Consequently, they were considered secondary factors in identifying prospective customer segments within the scope of this study.

6. Key Insight

The analysis generated several important insights regarding customer purchasing behaviour and the identification of prospective bike customers.

6.1 Middle-Aged Customers Represent the Primary Target Market

The Middle Age category (31–53 years) emerged as the strongest customer segment in the analysis. This group constituted approximately 70% of the total customer population and recorded the highest purchase rate of 54.6%. These findings indicate that middle-aged customers represent the company's primary target market for future marketing campaigns.

6.2 Income Influences Purchasing Behaviour but Is Not the Sole Determinant

The analysis revealed that customers with relatively higher average incomes generally exhibited stronger purchasing behaviour. However, the Old age group recorded the highest average income while producing the lowest purchase rate. Similarly, Clerical workers maintained a relatively high purchase rate despite earning below the overall average income. These findings suggest that although income contributes to purchasing behaviour, it should not be used as the sole criterion for customer targeting.

6.3 Region Provides Additional Market Opportunities

Regional analysis showed that all three geographical regions demonstrated market potential. Although the Pacific region recorded the highest purchase rate among middle-aged customers, North America contained the largest concentration of customers. This suggests that marketing strategies should consider both customer volume and purchase rate when allocating resources across regions.

6.4 Occupation Enhances Customer Segmentation

Occupation was identified as an important socioeconomic factor that complements age and income in explaining purchasing behaviour. Professional, Management, Skilled Manual, and Clerical occupations demonstrated favorable purchase rates, indicating that

occupation can improve the identification of prospective customers when combined with other customer characteristics.

6.5 Effective Marketing Requires Customer Segmentation

The findings demonstrate that customer purchasing behaviour varies across demographic, geographic, and socioeconomic characteristics. Consequently, a single marketing strategy may not effectively serve all customer groups. A segmented marketing approach is therefore more likely to improve customer engagement and increase bike sales.

7. RECOMMENDATIONS

Based on the findings of this analysis, the following recommendations are proposed:

1. Prioritize Middle-Aged Customers

Marketing campaigns should primarily target customers between 31 and 53 years of age, as this group demonstrated the highest purchase rate and represented the largest customer segment within the data set.

2. Implement Region-Specific Marketing Strategies

Marketing resources should be allocated according to regional performance. While the Pacific region demonstrated the highest purchase rate, North America represents a substantial customer base due to its larger population. Regional marketing campaigns should therefore be tailored to the characteristics of each market.

3. Combine Occupation with Income During Customer Targeting

Income should not be used as the only criterion for identifying prospective customers. Occupation should also be considered, as several occupational groups recorded strong purchase rates despite differences in average income.

4. Adopt a Customer Segmentation Strategy

Rather than implementing a uniform marketing approach, the company should segment customers according to age, region, occupation, and income. This will improve marketing efficiency and increase the likelihood of reaching high-potential customers.

5. Improve Future Data Collection

The data set used for this analysis was not evenly distributed across all customer groups. Some segments contained considerably fewer observations than others, reducing the reliability of conclusions for those categories. Future studies should aim to collect a more

balanced dataset to improve the accuracy, representativeness, and reliability of analytical findings.

8. LIMITATIONS OF THE STUDY

Although the analysis generated valuable insights, several limitations should be acknowledged.

The dataset used in this study consisted of 1,000 customer records obtained from a public GitHub repository. As a secondary dataset, the researcher had no control over the data collection process or the selection of respondents.

Furthermore, the customer segments were not evenly distributed across all variables. Some categories contained relatively small sample sizes, which may have influenced the reliability of comparisons and reduced confidence in conclusions relating to those groups.

Additionally, the analysis was limited to the variables available within the dataset. Other factors that may influence bike purchasing behaviour, such as customer preferences, product pricing, promotional activities, brand perception, and economic conditions, were not captured and therefore could not be evaluated.

9. CONCLUSION

This study analysed customer demographic, geographic, and socioeconomic characteristics to identify prospective bike customer segments for targeted marketing.

The findings revealed that age was the most influential demographic characteristic, with middle-aged customers exhibiting both the largest customer population and the highest purchase rate. Further analysis demonstrated that region, occupation, and income also contributed to customer purchasing behaviour, although income alone was not sufficient to explain purchasing decisions.

Overall, the analysis highlights the importance of customer segmentation in supporting effective marketing strategies. By focusing on high-potential customer groups and adopting data-driven decision-making, the company can improve marketing efficiency, optimize resource allocation, and enhance future sales performance.